MA3521 — In-Class Worksheet Tuesday/Wednesday
Do not solve.

Name:

1. ☐ Linear ☐ Separable $\quad ty' + 2y = e^t \qquad y(1) = 0$

Show a suitable intermediate step

Show a fully integrated answer

2. ☐ Linear ☐ Separable $\quad y' = 2t\cos(t^2)(1 + y^2) \qquad y(\sqrt{\pi}) = 0$

Show a suitable intermediate step

Show a fully integrated answer

3. ☐ Linear ☐ Separable $\ln(y)y' = te^{t^2}$ $y(1) = e$

Show a suitable intermediate step

Show a fully integrated answer

4. ☐ Linear ☐ Separable $ty' + (t+1)y = 1$ $y(1) = 1$

Show a suitable intermediate step

Show a fully integrated answer